

Curriculum Vitae

Anargha Mondal

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EDUCATION

Indian Institute of Science Education and Research Pune
Bachelor of Science-Master of Science (BS-MS), **Physics Major**

Expected Graduation: August 2027

CGPA: 8.6/10

RESEARCH EXPERIENCE

Master's Thesis, IISER Pune

Maharashtra, India

May 2026 - Present

- Supervised by Dr Arka Banerjee and in collaboration with Dr Adrian Bayer. The goal of the project is to investigate the imprints of scale dependence of the halo bias induced by massive neutrinos using weak lensing observables and the electron pressure in halos.
- The scale dependence is amplified by splitting the halo sample on various combinations of the projected mass profile and the tSZ signal in the halo environment, thus investigating a novel approach to constrain neutrino mass from future X-ray and lensing observations of galaxy clusters.

Semester Project, IISER Pune

Maharashtra, India

August 2024 - April 2026

- Supervised by Dr Arka Banerjee. The goal of the project was to compare constraints on cosmological parameters calculated using nearest neighbour statistics and the two-point correlation function.
- Calculated cross-correlations between the positions of the top 10^5 most massive halos and the matter density field with the Quijote simulations for two different statistics for the tracer $\times$ field formalism: the two-point correlation function and the k -nearest neighbour cumulative distribution function.
- Derived Fisher forecasts for the two statistics and compared the constraints for cosmological parameters. The k -NN CDFs consistently show better constraints than the two-point correlation function.
- Codes written during this project and some other one were compiled into the *kNNpy* Python package.

Research Project, NAOJ, Mitaka

Hybrid

May 2024 - February 2025

- Supervised by Dr Maria Giovanna Dainotti. The goal of the project was to review the status of the Hubble Tension based on the latest Type Ia SNe observations.
- Developed a new duplicate-free combination of Type Ia supernovae from Pantheon Plus, Pantheon, DES, JLA and SH0ES, called the 'Master sample'.
- Applied three different binning techniques to the Master sample, and chose two different dark matter models - Λ CDM and w_0w_a CDM - to investigate the effect of the binning technique and the equation of state of dark energy on the Hubble constant.
- Performed a Markov Chain Monte Carlo analysis of the binned Master sample to determine the H_0 value for each redshift bin, for various Dark Energy models. Fixed various parameters to study its effect on the Hubble constant and its effect on the Hubble tension.
- Results indicate a slow redshift-evolution towards lower values of H_0 .
- Published in the *Journal of High Energy Astrophysics (JHEAP)*.

Reading Project, TIFR Mumbai

Online

July 2023 - December 2023

- Supervised by Dr. Sudip Bhattacharyya. The reading project concerned accretion processes in compact object binaries, with understanding their applications in real data.
- Tested how well our current theoretical models of accretion disks in Black Hole Binaries and Low Mass X-Ray Binaries fit observed data, using data from the Large X-ray Photon Counter and Soft X-ray Telescope on the AstroSat.

SOFTWARE DEVELOPMENT

kNNpy

May 2025 - Present

- A Python package for higher-order cosmological clustering using nearest-neighbour distributions. Links to the website and the GitHub repository
- Contributed towards building the codes, ideation and design.
- Active maintainer and contributor of the package and website, currently working on future versions.

COMPUTATIONAL SKILLS

Languages: Python, Bash

Tools: Mathematica, MATLAB, Git and GitHub, L^AT_EX, HEASoft

PUBLICATIONS

- ***A New Master Supernovae Ia sample and the investigation of the H_0 tension***
Dainotti et al, including Mondal, A. (2025). (JHEAP)

AWARDS & FELLOWSHIPS

- National Top 1% in National Graduate Physics Examination (NGPE) *February, 2023*
- Kishore Vaigyanik Protsahan Yojana (KVPY) Fellow *August, 2021*
- National Talent Search Examination (NTSE) Scholar *March, 2021*
- Indian Olympiad Qualifier in Mathematics (IOQM) Merit Awardee *January, 2021*
- Jagadish Bose National Talent Search Scholar *December, 2020*

TALKS & CONFERENCES

Talks

- **“Investigating the Hubble Tension with Type Ia Supernovae”** *14th November, 2024*
Contributed Talk, Lagrangians to Lasers, IISER Pune, India.
- **“Investigating the Hubble Tension with Type Ia Supernovae”** *13th November, 2024*
Invited Talk, Science Club, IISER Pune, India.

Conferences

- **Pune-Mumbai Cosmology and Astro-Particle Meeting (PMCAP)** *27th – 29th March, 2026*
Sixth edition organised at NCRA, Pune, Maharashtra, India
- **Pune-Mumbai Cosmology and Astro-Particle Meeting (PMCAP)** *12th – 14th September, 2025*
Fifth edition organised at IISER Pune, Maharashtra, India
- **Pune-Mumbai Cosmology and Astro-Particle Meeting (PMCAP)** *7th – 8th February, 2025*
Fourth edition organised at IIT Bombay, Maharashtra, India
- **Conference on Blazars and Restless AGN (COBRA)** *22nd July, 2024*
Organised by IUCAA, Pune and Presidency University, Kolkata

CO-SCHOLASTIC ACTIVITIES

- Coordinator, Helicase, the Science Magazine of IISER Pune (March 2023 - February 2024)
- Physics Question-Making Team Member, Mimamsa (October 2022 - Present)
- ‘Cosmic Forums’ Team Member, Aakashganga, Astronomy Club of IISER Pune
- Regular speaker in the Lagrangians to Lasers and Cosmology Journal Clubs
- Core Team Member, Math Club, IISER Pune